

MAT1575, Classwork5, Fall2026

1.5 Substitution (p.74-81)

1. Evaluate each of the following indefinite integrals:

(a) $\int (x^2 - 3)^2 dx$

(b) $\int (x^2 - 3)^3 2x dx$?

$$\int (x^4 - 6x^2 + 9) dx = \frac{x^5}{5} - 2x^3 + 9x + C$$

Observation: Let $g(x) = x^2 - 3$. Then $(x^2 - 3)^3 \cdot 2x = [g(x)]^3 \cdot g'(x)$. Can it be used to get the results in an easier way?

2. **Theorem: u-Substitution or Change of Variables** with Indefinite Integrals.

Let $u = g(x)$, where $g'(x)$ is continuous over an interval, let $f(x)$ be continuous over the corresponding range of g , and let $F(x)$ be an antiderivative of $f(x)$. Then

$$\int f[g(x)]g'(x)dx = \int f(u)du = F(u) + c = F(g(x)) + c.$$

Proof: Since $\frac{d}{dx} F(g(x)) =$, then $\int f(g(x))g'(x)dx =$

Let . Then and

Note: what does $du = g'(x)dx$ mean?

$$\Delta x = x_i - x_{i-1} \text{ and } \Delta u = g(x_i) - g(x_{i-1}). \quad \frac{\Delta u}{\Delta x} = \text{---} \Rightarrow \Rightarrow$$

3. Using u-Substitution to Find an Antiderivative of the given indefinite integrals:

(a) $\int (x^2 - 3)^3 2x dx$ $u = x^2 - 3$
 $du = 2x \cdot dx$

(b) $\int 6y(3y^2 + 4)^4 dy$

(c) $\int z\sqrt{z^2 - 5} dz$

$$\int u^3 du = \frac{u^4}{4} + C = \frac{(x^2 - 3)^4}{4} + C$$

4. **Problem-Solving Strategy of Integration by Substitution.**

- (1) _____
- (2) _____
- (3) _____
- (4) _____

5. Using u-Substitution to Find an Antiderivative of the given indefinite integrals:

(a) $\int \frac{\sin t}{\cos^3 t} dt$

(b) $\int \cos^3 \theta \sin \theta d\theta$

(c) $\int \frac{x}{\sqrt{x-1}} dx$

6. **Theorem: u-Substitution or Change of Variables** with Definite Integrals.

Let $u = g(x)$, where $g'(x)$ is continuous over an interval, let $f(x)$ be continuous over the corresponding range of g , and let $F(x)$ be an antiderivative of $f(x)$. Then

$$\int_a^b f(g(x))g'(x)dx = \int_{g(a)}^{g(b)} f(u)du.$$

Observation: _____

7. Using u-Substitution to Find an Antiderivative of the given definite integrals:

$$(a) \int_0^1 x^2(1 + 2x^3)^5 dx$$

$$(b) \int_{-1}^0 y(2y^2 - 3)^5 dy$$

$$(c) \int_0^1 xe^{4x^2+3} dx$$

8. Using u-Substitution to Find an Antiderivative of the given definite integrals:

$$(a) \int_0^1 x^2 \cos\left(\frac{\pi}{2}x^3\right) dx$$

$$(b) \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta$$

$$(c) \int_0^{\frac{\pi}{4}} \sec^2 \theta \tan \theta d\theta$$